

Project Title

SpecMind AI: AI-Driven Software Requirements Analysis and Validation System

Abstract

Software projects often fail because of unclear, incomplete, or inconsistent requirements identified too late in the development process. These issues lead to delays, increased costs, and poor software quality. To address this challenge, **SpecMind AI** is proposed as an intelligent platform that helps software teams improve requirement quality before development begins.

The platform uses **Artificial Intelligence (AI)** to analyze **Software Requirement Specification (SRS)** documents or user-provided requirements. It automatically identifies ambiguous, incomplete, duplicate, and conflicting requirements, while suggesting clearer and more precise alternatives. In addition, the system predicts potential project risks based on the quality of the requirements and provides an overall requirement quality score.

To support Agile software development, SpecMind AI automatically generates user stories, acceptance criteria, prioritizes requirements, estimates story points, and creates an initial product backlog and sprint plan. Interactive dashboards provide clear insights into requirement quality, project risks, and development readiness, enabling teams to make informed decisions at an early stage.

By integrating AI into the requirements engineering process, SpecMind AI reduces manual effort, improves communication between stakeholders and developers, minimizes requirement-related errors, and increases the chances of successful software delivery. The platform offers a modern, practical, and intelligent approach to software requirements management, making it well suited for today's AI-driven software development environment.